

Time Open Ended Assignment

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SINGLE VARIABLE - NON-STANDARD FORMATTING

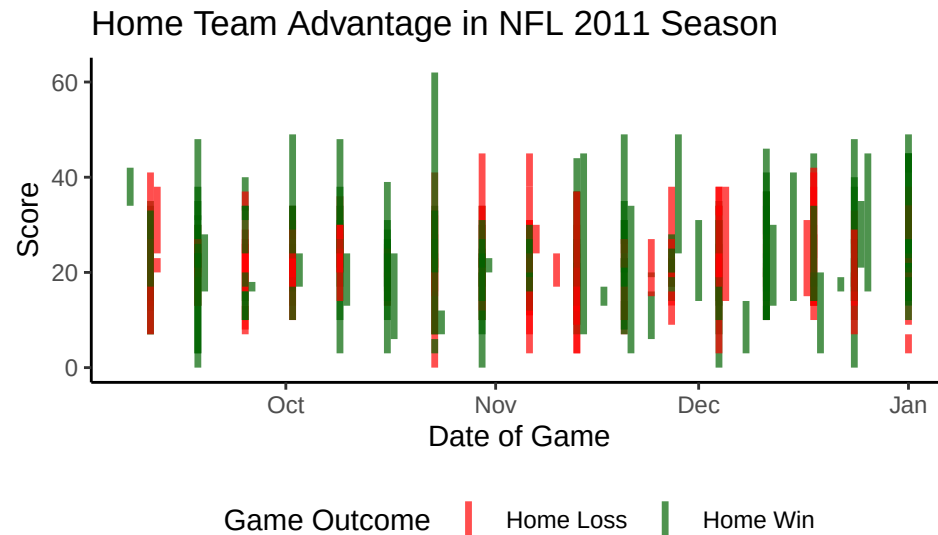
Do home teams have an advantage in scoring?

```
data(NFLScores2011)
head(NFLScores2011)
```

##	Week	HomeTeam	AwayTeam	HomeScore	AwayScore	HomeYards
## 1	1	Green Bay Packers	New Orleans Saints	42	34	399
## 2	1	Chicago Bears	Atlanta Falcons	30	12	377
## 3	1	Jacksonville Jaguars	Tennessee Titans	16	14	323
## 4	1	Tampa Bay Buccaneers	Detroit Lions	20	27	315
## 5	1	St. Louis Rams	Philadelphia Eagles	13	31	335
## 6	1	Baltimore Ravens	Pittsburgh Steelers	35	7	385

##	AwayYards	HomeTO	AwayTO	Date	Day
## 1	477	0	1	8-Sep	Thu
## 2	386	1	3	11-Sep	Sun
## 3	292	1	1	11-Sep	Sun
## 4	431	2	1	11-Sep	Sun
## 5	403	1	1	11-Sep	Sun
## 6	312	0	7	11-Sep	Sun

```
NFL_clean <- NFLScores2011 %>%mutate(Year = ifelse(Week == 17, 2012, 2011), NewDate = dmy(paste0(Date, Year)))
gf_linerange(HomeScore + AwayScore ~ NewDate, data = NFL_clean, color = ~HomeWin, alpha = 0.7, size = 1)
```



MULTIPLE VARIABLES - ALL INDIVIDUAL PARTS

How does storm wind speed vary by storm category over time?

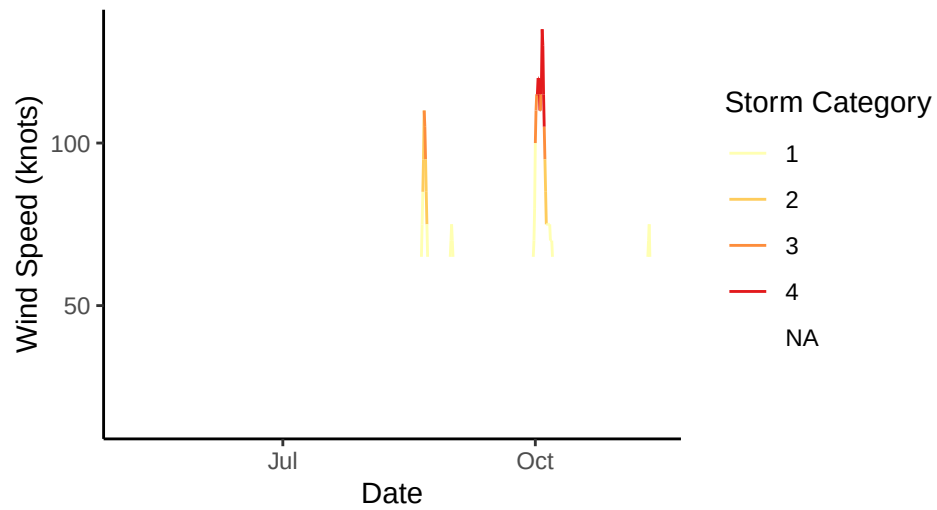
```
data(storms)
head(storms)
```

```
## # A tibble: 6 x 13
##   name   year month   day  hour   lat   long status   category  wind pressure
##   <chr> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <fct>      <dbl> <int>    <int>
## 1 Amy    1975     6    27     0  27.5 -79 tropical de~    NA     25     1013
## 2 Amy    1975     6    27     6  28.5 -79 tropical de~    NA     25     1013
## 3 Amy    1975     6    27    12  29.5 -79 tropical de~    NA     25     1013
## 4 Amy    1975     6    27    18  30.5 -79 tropical de~    NA     25     1013
## 5 Amy    1975     6    28     0  31.5 -78.8 tropical de~    NA     25     1012
## 6 Amy    1975     6    28     6  32.4 -78.7 tropical de~    NA     25     1012
## # i 2 more variables: tropicalstorm_force_diameter <int>,
## #   hurricane_force_diameter <int>
```

```
storms_clean <- storms %>% mutate(DateTime = make_datetime(year = year, month = month, day = day, hour = hour))
storms_2015 <- storms_clean %>% filter(year == 2015)
gf_line(wind ~ DateTime, data = storms_2015, color = ~Category, group = ~name) %>% gf_refine(scale_color = ~Category)
```

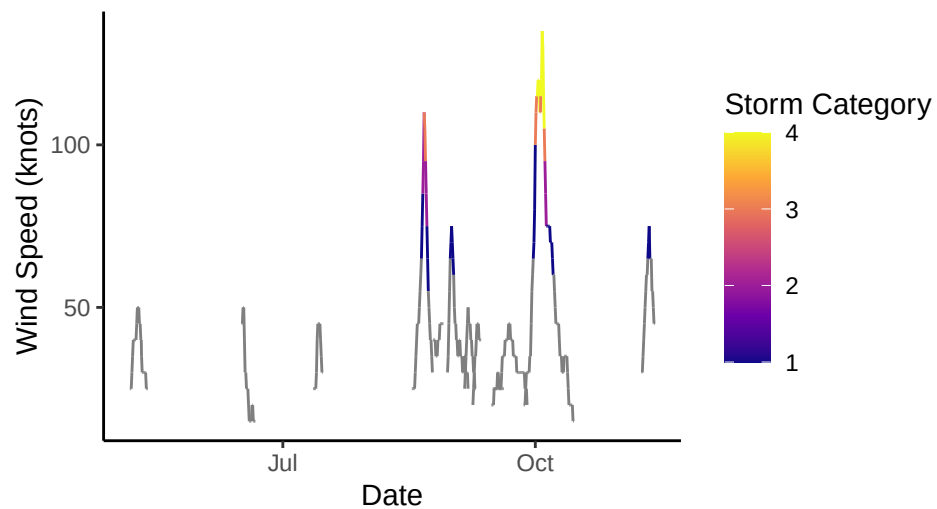
```
## Warning: Removed 289 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

Storm Wind Speeds by Category – 2015



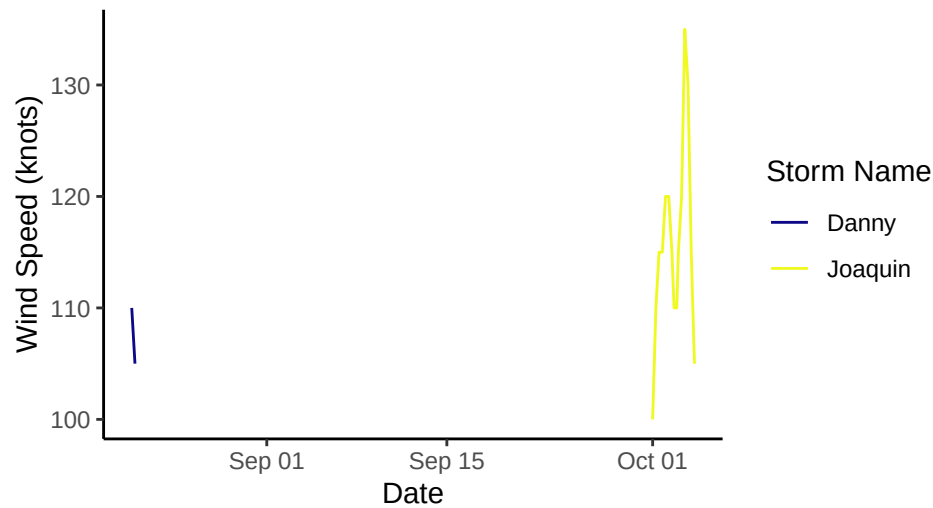
```
gf_line(wind ~ DateTime, data = storms_2015, color = ~category, group = ~name) %>%gf_refine(scale_color_viridis)
```

Storm Wind Speeds by Category – 2015



```
storms_major <- storms_2015 %>% filter(category >= 3)
gf_line(wind ~ DateTime, data = storms_major, color = ~name, group = ~name) %>%gf_refine(scale_color_viridis)
```

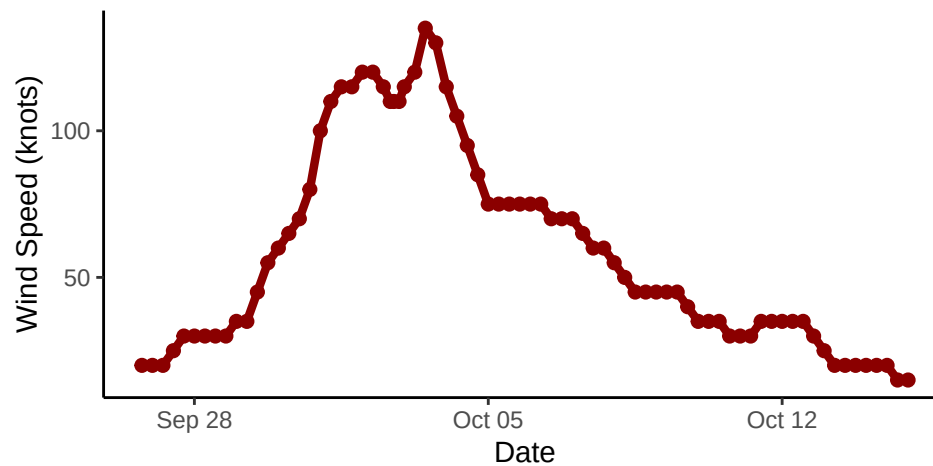
Major Hurricane Wind Speeds – 2015



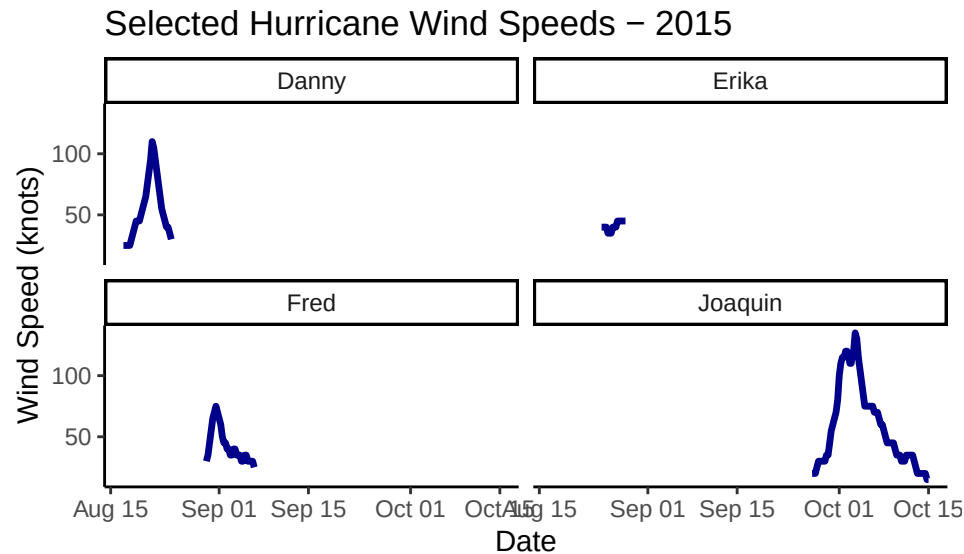
```
storm_example <- storms_2015 %>% filter(name == "Joaquin")
gf_line(wind ~ DateTime, data = storm_example, color = "darkred", size = 1.5) %>% gf_point(wind ~ DateTime)
```

Hurricane Joaquin Wind Speed – 2015

Wind speed progression over time



```
storms_selected <- storms_2015 %>% filter(name %in% c("Joaquin", "Danny", "Erika", "Fred"))
gf_line(wind ~ DateTime | name, data = storms_selected, color = "darkblue", size = 1.2) %>% gf_labs(title)
```



ADDITIONALS DIFFICULTIES - NON-STANDARD MONTH SECTION

What are the seasonal patterns in retail sales?

```
data(RetailSales2011)
head(RetailSales2011)
```

```
##   Month Year Sales
## 1   Jan 2000 294.6
## 2   Feb 2000 252.5
## 3   Mar 2000 245.5
## 4   Apr 2000 244.0
## 5   May 2000 257.9
## 6   June 2000 244.9
```

```
RetailSales_clean <- RetailSales2011 %>% mutate(Month.Num = ifelse(str_length(Month) == 3, match(Month, month.abb), 0))
head(RetailSales_clean)
```

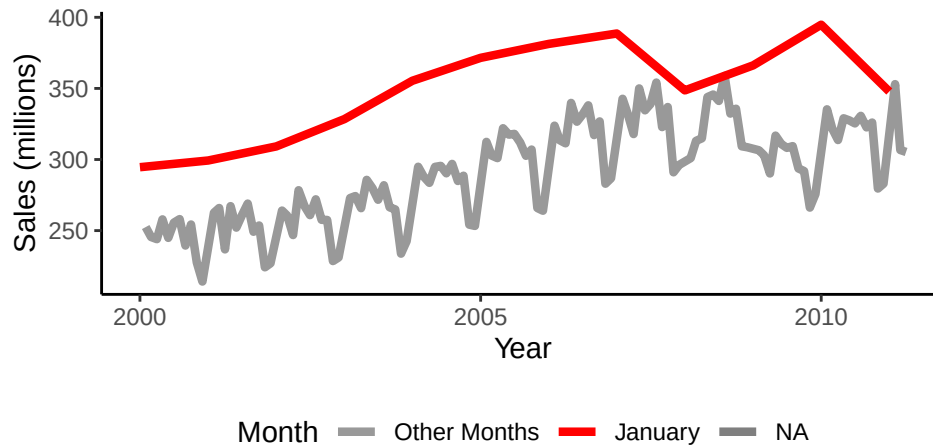
```
##   Month Year Sales Month.Num      Date Quarter IsJanuary
## 1   Jan 2000 294.6         1 2000-01-01         1      TRUE
## 2   Feb 2000 252.5         2 2000-02-01         1     FALSE
## 3   Mar 2000 245.5         3 2000-03-01         1     FALSE
## 4   Apr 2000 244.0         4 2000-04-01         2     FALSE
## 5   May 2000 257.9         5 2000-05-01         2     FALSE
## 6   June 2000 244.9        6 2000-06-01         2     FALSE
```

```
gf_line(Sales ~ Date, data = RetailSales_clean, color = ~IsJanuary, size = 1.5)%>%gf_refine(scale_color = "discrete")
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

Retail Sales Show Consistent January Peaks

January consistently shows highest sales each year



```
gf_boxplot(Sales ~ factor(Quarter), data = RetailSales_clean, fill = ~factor(Quarter)) %>%gf_refine(sca
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

Retail Sales Distribution by Quarter

Quarterly sales patterns 2000–2011

